

Projecting possible future trajectories for systems of cities

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Special Session: Validation of Spatial Simulation Models

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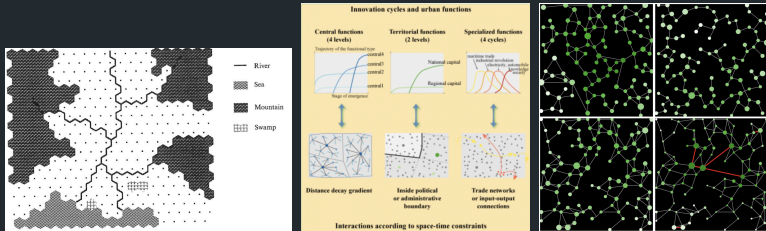
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Simulating systems of cities [Pumain, 2011], with the perspective of designing multi-level policies [Rozenblat and Pumain, 2018]: example of Simpop1 [Sanders et al., 1997], Simpop2 [Pumain, 2011], SimpopNet [Schmitt, 2014]

Many advanced methods to explore and validate simulation models [Raimbault and Pumain, 2019]:

- Calibration with distributed genetic algorithms [Schmitt et al., 2015]
- Calibration profile along one dimension [Reuillon et al., 2015]
- Simultaneous calibration of multiple model using niched GAs [Cottineau et al., 2015]
- Diversity search to explore feasible output space [Chérel et al., 2015]
- Synthetic spatial configurations [Raimbault, 2019][Raimbault, 2020b]
- Sensitivity to the spatial configuration [Raimbault et al., 2019]
- More in development: inverse problem, probabilistic diversity search, participatory validation, . . .

All implemented into the OpenMOLE platform [Reuillon et al., 2013]



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<https://exmodelo.org/>

→ need for methods to project *potential future trajectories* of systems of cities

→ in a context of urban system dynamics and sustainability: trade-offs between Sustainable Development Goals

[Raimbault and Pumain, 2022], among 5 dimensions of SDG indicators [Raimbault and Pumain, 2024]

Research objective:

First tests to combine exploration methods to explore future trajectories in a multi-model of simulation for systems of cities.

Combination of existing model exploration methods to project future potential trajectories for a system of cities; given a simulation model which can be calibrated on real data:

1. Calibrate the model with multiple objectives using NSGA2 [Schmitt et al., 2015], to obtain a set of solutions on a Pareto front (for example population fit errors used by [Raimbault, 2020a]);
2. extract the statistical distribution of model parameters from the Pareto front and establish probable boundaries for each;
3. explore the model output space with the PSE diversity search algorithm [Chérel et al., 2015], on a few time steps in the future, using previously obtained parameter bounds.

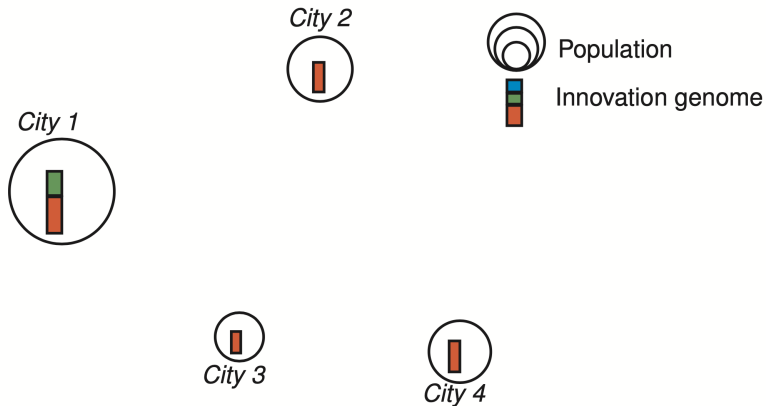
Models integrated by [Raimbault and Pumain, 2024] (benchmarked by [Raimbault et al., 2020a] on several large urban systems):

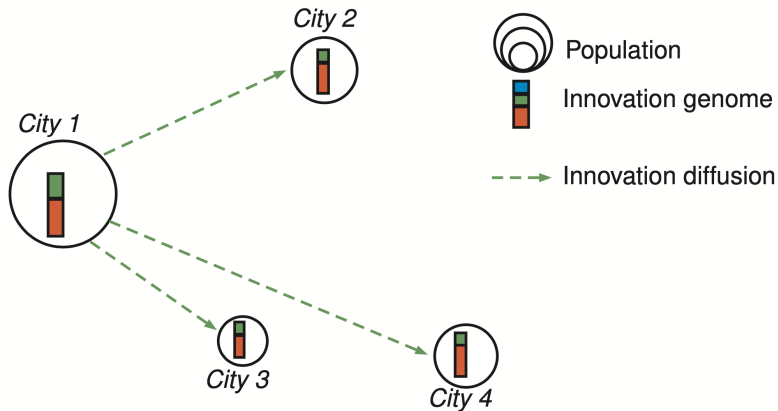
- Innovation diffusion urban evolution model [Raimbault and Pumain, 2022]
- Marius model for economic exchanges [Cottineau et al., 2015]
- Co-evolution model for cities and infrastructure networks [Raimbault, 2021]

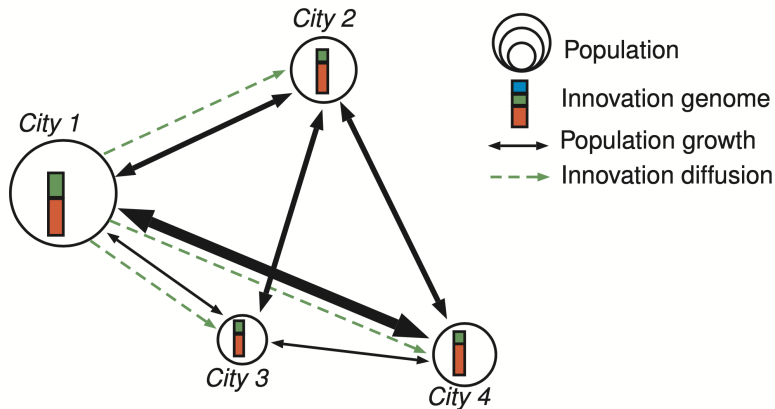
“Semi-weak” coupling of submodels:

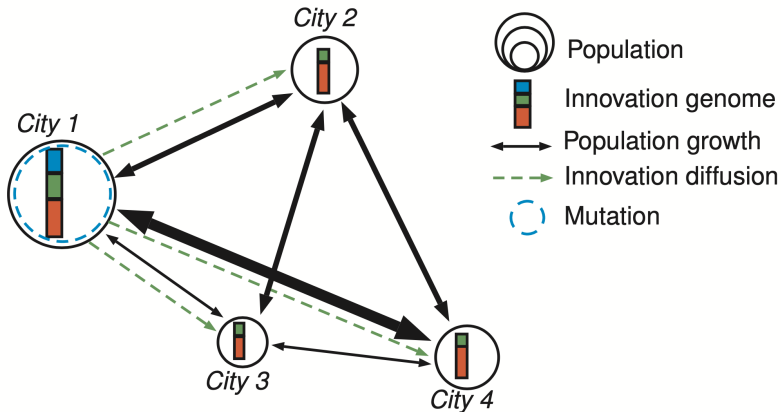
$$\begin{aligned} S_0 \rightarrow \left[S_1^{(0)} = M_0(S_0) \rightarrow \dots \rightarrow S_1^{(K-1)} = M_{K-1}(S_1^{(K-2)}) = S_1 \right] \longrightarrow \dots \\ \longrightarrow \left[S_T^{(0)} = M_0(S_{T-1}) \rightarrow \dots \rightarrow S_T^{(K-1)} = M_{K-1}(S_T^{(K-2)}) = S_T \right] \quad (1) \end{aligned}$$

Remarks: strictly weak coupling does not capture dynamics nor synergies; stronger coupling may exist to better capture interdependencies processes (here no specific coupling ontology, only population and distance matrix are shared between submodels)









At each time step, with $P_i(t)$ population, $\delta_{c,i}(t)$ genome, u_c utility of innovation, $p_{c,i,t}$ share of total population adopting innovation c in city i

1. Crossover through the diffusion of innovations

$$\delta_{c,i,t} = \frac{\sum_j p_{c,j,t-1}^{\frac{1}{u_c}} \cdot \exp\left(-\frac{d_{ij}}{d_i}\right)}{\sum_c \sum_j p_{c,j,t-1}^{\frac{1}{u_c}} \cdot \exp\left(-\frac{d_{ij}}{d_i}\right)}$$

2. Population growth through spatial interactions

$$P_i(t) - P_i(t-1) = w_i \cdot \sum_j \frac{V_{ij}}{\langle V_{ij} \rangle} \text{ with}$$

$$V_{ij} = \frac{P_i(t-1) \cdot P_j(t-1)}{(\sum_k P_k(t-1))^2} \cdot \exp\left(-\frac{d_{ij}}{d_G}\right) \cdot \prod_c \delta_{c,i,t}^{\phi_{c,t}}$$

$$\text{and } \phi_{c,t} = \sum_i \delta_{i,c,t} \cdot P_i(t-1) / \sum_i P_i(t-1)$$

3. Mutations with innovations introduced with probability

$\beta \cdot (P_i(t) / \max_k P_k(t))^{\alpha_I}$ and an initial penetration rate r_0 ; new utility u_c randomly distributed (normal or log-normal) with average current average utility and standard deviation a given parameter σ_U

MARIUS family of models [Cottineau et al., 2015]:

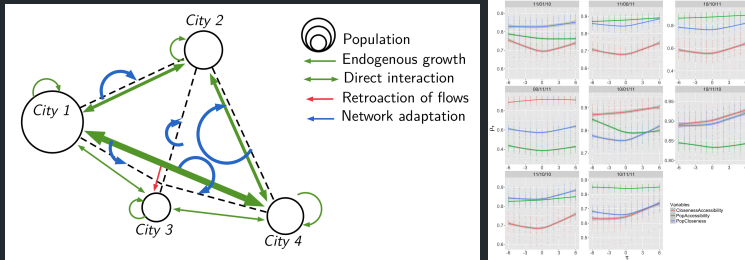
Initial wealth as a power law of population (exponent α_W)

1) Update supply and demands as super-linear functions of population (exponents α_S, α_D)

2) Exchange goods according to a gravity potential of interaction (distance decay d_M), supplies and demands; update wealth accordingly

3) Update population such that population difference is a power law of wealth difference (economic multiplier e_M and exponent α_P)

System of cities interaction model including network evolution.



Raimbault, J. (2020). Indirect evidence of network effects in a system of cities. *Environment and Planning B: Urban Analytics and City Science*, 47(1), 138-155.

Raimbault, J. (2021). Modeling the co-evolution of cities and networks. In *Handbook of Cities and Networks* (pp. 166-193). Edward Elgar Publishing.

Raimbault, J. (2025). Hierarchy and co-evolution processes in urban systems. *Hierarchy in infrastructure networks*, J. Fen-Chong, ed. ISTE Editions.

Model implemented in `scala`; relatively large parameter space; integration into the `spatialdata scala` library for spatial sensitivity analysis [Raimbault et al., 2020b]

→ integration into the OpenMOLE model exploration open source software [Reuillon et al., 2013]

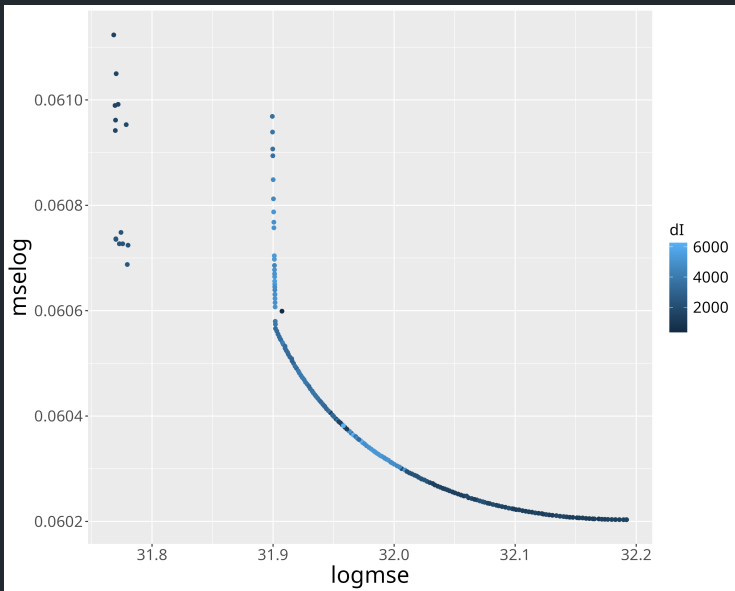


Enables seamlessly (i) model embedding; (ii) access to HPC resources; (iii) exploration and optimization algorithms

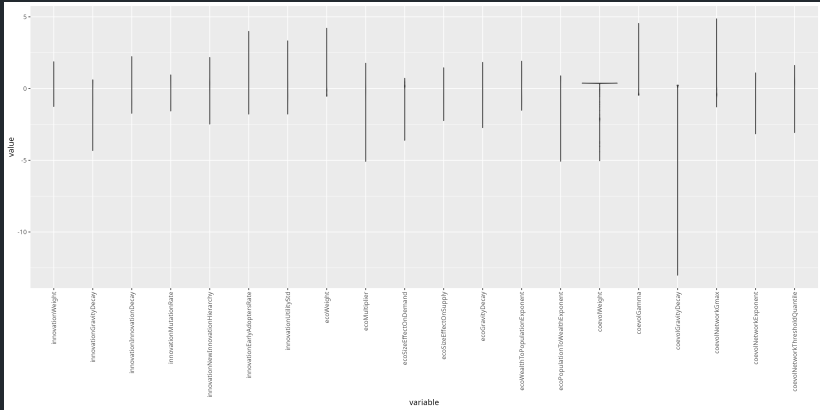
<https://openmole.org/>

- Apply the method to the multi-model, on Europe and China with the GeoDiversity harmonised data ([Cura et al., 2017], and the ChinaCities database [Swerts, 2017])
- Calibrate on population trajectories only (Mean Square Error on log of populations and log of Mean Square Error), $\simeq 1960 - 2010$; optimisation parameters: 8 parameters for innovation, 7 parameters for economic exchanges, 6 parameters for co-evolution;
- Project trajectories on 3 time steps after 2010, until 2040
- PSE objectives (6 dimensions): average of $\log(\text{population})$ (2020, 2030, 2040) and standard deviation of $\log(\text{population})$ (2020, 2030, 2040);
- Compute SDGs indicators for the projected trajectories: total utility of innovations (SDG 9 “Innovation”, total spatial interaction flows across submodels (SDG 13 “Climate”), average distance between cities (SDG 9 “Resilient infrastructure”); economic inequalities (SDG 10 “Inequalities”); total wealth of cities (SDG 8 “Economic Growth”).

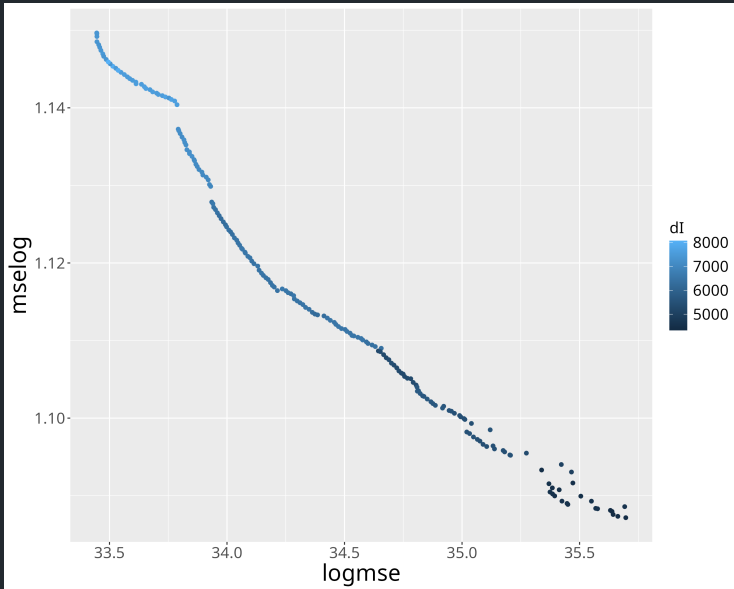
Results: EU calibration



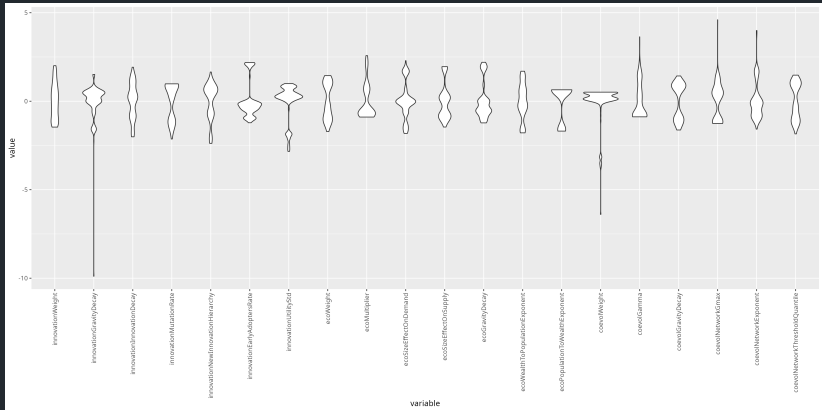
Results: EU calibration parameters



Results: ZH calibration

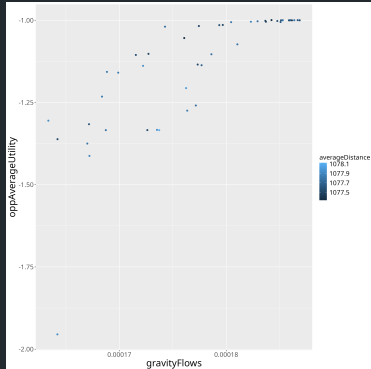
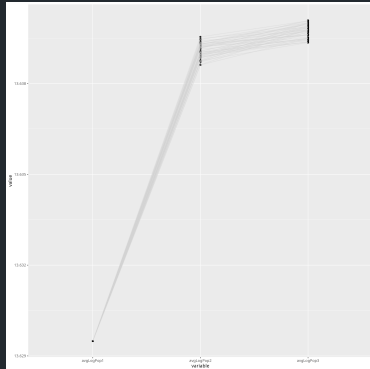


Results: ZH calibration parameters

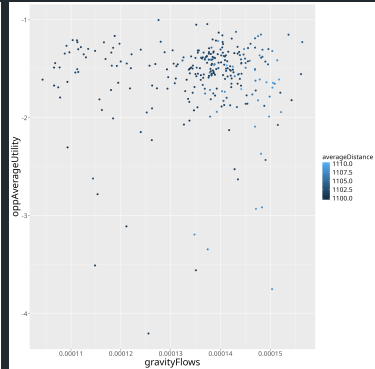
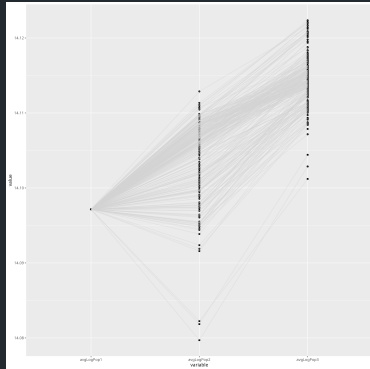


→ broader bounds for ZH (1 order of magnitude for innovationn and economic weight for example), meaning much broader potential futures.

Results: PSE on projected trajectories for EU



Results: PSE on projected trajectories for ZH



→ confirmed much broader feasible trajectory space for Chinese urban system

Contributions:

- A new method to project possible future urban systems trajectories
- First calibration of the multi-model of simulation for urban systems sustainability
- Under the stationarity assumption, broader urban futures for China vs EU

Perspectives:

- Calibration on other dimensions (e.g. innovation and emissions) for robustness of SDGs trajectories
- Stronger coupling between submodels
- Test of the method on simpler simulation models and benchmark with similar methods

Code and data:

Model <https://github.com/openmole/spatialdata>

Method and results <https://github.com/JusteRaimbault/SDGTradoffs>



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